22. Word Codes (Set B)

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Encoding with a given pair:

In a code, the word LAMP is written as NCOR. Work out the shift being used, then use it to encode the word 'DESK'.

Answer: L→N, A→C, M→O, P→R: each letter moves forward 2 places, so the shift is +2. Applying +2 to DESK: D→F, E→G, S→U, K→M. Answer: **+2, so DESK becomes FGUM**.

Example 2 — Decoding with a stated shift:

In a code, every letter is shifted BACK by 3 places (D→A, E→B, F→C…). What does the code word 'ZLQG' decode to?

Answer: Shift each letter back 3 places: Z→W, L→I, Q→N, G→D. Answer: **WIND**.

Example 3 — Multiple-choice:

Using the LAMP → NCOR code (shift +2), what is the code for SHED? A) UJGF B) UJGE C) TJGF D) UIGF

Answer: S→U, H→J, E→G, D→F. Answer: **A) UJGF**.

Example 4 — Select-all:

Select ALL the words below that are correctly encoded using the LAMP → NCOR code (shift +2). A) FISH → HKUJ B) BOOK → DQQM C) DESK → FGUN D) ROPE → TQRG E) TREE → VTFG

Answer: FISH→HKUJ, BOOK→DQQM and ROPE→TQRG are all shifted correctly forward 2 places, so **A, B and D** are correct. DESK should become FGUM (not FGUN), and TREE should become VTGG (not VTFG).

Example 5 — A code that wraps around:

Using a +4 shift (where the alphabet wraps round after Z), what is the code for YOGA?

Answer: Y→C (wrapping round past Z), O→S, G→K, A→E. Answer: **CSKE**.

Foundation

15 questions

1. Using the LAMP → NCOR code (shift +2), what is the code for SHIP?

2. Using the LAMP → NCOR code (shift +2), what is the code for DESK?

3. Using the LAMP → NCOR code (shift +2), what is the code for MILK?

4. Using the LAMP → NCOR code (shift +2), what is the code for BOOK?

5. Using the LAMP → NCOR code (shift +2), what is the code for CORN?

6. Using the LAMP → NCOR code (shift +2), what is the code for RUG?

7. Using the LAMP → NCOR code (shift +2), what is the code for HIVE?

8. Using the LAMP → NCOR code (shift +2), what is the code for BELL?

9. Using the LAMP → NCOR code (shift +2), what is the code for NEST?

10. Using the LAMP → NCOR code (shift +2), what is the code for SAND?

11. Using the LAMP → NCOR code (shift +2), what is the code for LEAF?

12. Using the LAMP → NCOR code (shift +2), what is the code for CORK?

13. Using the LAMP → NCOR code (shift +2), what is the code for SOAP?

14. Using the LAMP → NCOR code (shift +2), what is the code for SHED? A) UJGF B) UJGE C) TJGF D) UIGF

15. Select ALL the words below that are correctly encoded using the LAMP → NCOR code (shift +2). A) FISH → HKUJ B) BOOK → DQQM C) DESK → FGUN D) ROPE → TQRG E) TREE → VTFG

Standard

20 questions

1. Using the CRAB → GVEF code (shift +4), what is the code for FROST?

2. Using the CRAB → GVEF code (shift +4), what is the code for CHILD?

3. Using the CRAB → GVEF code (shift +4), what is the code for GRAPE?

4. Using the CRAB → GVEF code (shift +4), what is the code for STAGE?

5. Using the CRAB → GVEF code (shift +4), what is the code for STONE?

6. Using the CRAB → GVEF code (shift +4), what is the code for CLOCK?

7. Using the CRAB → GVEF code (shift +4), what is the code for SMILE?

8. Using the CRAB → GVEF code (shift +4), what is the code for TRAIN?

9. Using the CRAB → GVEF code (shift +4), what is the code for QUEEN?

10. Using the CRAB → GVEF code (shift +4), what is the code for SHINE?

11. Using the CRAB → GVEF code (shift +4), what is the code for SHELF? A) WLIPJ B) WLIOJ C) VLIPJ D) WLIQJ

12. Select ALL the words below that are correctly encoded using the CRAB → GVEF code (shift +4). A) WHEAT → ALIEX B) BENCH → FIRGK C) DRESS → HVIWW D) JUICE → NYMGH E) PRIZE → TVMDI

13. In a code, the word MOON is written as RTTS. Work out the shift being used, then use it to encode the word ‘STAR’.

14. In a code, the word BIKE is written as ELNH. Work out the shift being used, then decode the word ‘KHDW’ using the same shift.

15. In a code, every letter is shifted BACK by 4 places (E→A, F→B, G→C…). What does the code word ‘JMVI’ decode to?

16. In a code, every letter is shifted BACK by 5 places (F→A, G→B, H→C…). What does the code word ‘LQTB’ decode to?

17. In a code, every letter is shifted forward by 6 places (A→G, B→H…). Encode the word ‘MIST’.

18. Using a +6 shift code, what word does the code ‘RGQK’ represent?

19. In a code, every letter is shifted BACK by 2 places (C→A, D→B…). What does the code word ‘YKPF’ decode to? A) WIND B) VIND C) WINE D) XIND

20. In a code, the word CLAY is written as FODB. Work out the shift being used, then decode the word ‘VLQJ’ using the same shift.

Hard

5 questions

1. In a code, each letter is replaced by the letter that is the same number of places from the END of the alphabet as it is from the START (so A ↔ Z, B ↔ Y, C ↔ X…). What does the word ‘FISH’ become under this rule?

2. In a code, vowels (A, E, I, O, U) are shifted forward by 2, but consonants stay the same. Encode the word ‘SNAKE’ using this rule.

3. A two-step code first reverses the order of the letters in the word, then shifts every letter forward by 2 places. Apply this to the word ‘FISH’ and give the final result.

4. In a code, each letter's position number has 6 added to it, then it's converted back to a letter. What does the letter S become under this rule? A) Y B) X C) Z D) W

5. A puzzle gives the pair QUIZ → UYMD as a worked example (notice how Z wraps round to D). Work out the shift being used, then use it to encode the word ‘YOGA’ with the same shift.

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